

Findings and Discussion

Introduction

This chapter reports the empirical findings of the dual-evaluation framework introduced in Chapter 3. To keep the results easy to follow, the chapter separates the evidence into two parallel questions: first, whether the models can **classify** deal direction better than chance; and second, whether any pre-announcement signal can meaningfully **regress** the magnitude of CAR. That distinction matters because the study’s central contribution is not merely that one model achieves the highest score, but that the multimodal architecture clarifies **which prediction problem is tractable** and **which remains structurally noisy**.

The chapter therefore proceeds in a deliberately simple order. Section 4.2 reports the classification ablation ladder, because this is where the clearest empirical gains appear. Section 4.3 then reports the regression pipeline honestly, showing why continuous CAR magnitude remains difficult to predict. Section 4.4 resolves the M2 reversal and explains why naive NLP degrades rather than improves prediction. Sections 4.5 to 4.7 test H1, H2, and H3 directly. Section 4.8 presents interpretability evidence from SHAP, and Section 4.9 translates the classifier gain into practical financial meaning before closing with limitations.

All reported classification results derive from five-fold stratified cross-validation. All preprocessing steps — median imputation, scaling, and any trainable transformations — were fit on training folds only, then applied to the held-out fold, preserving the leakage controls established in Chapter 3 [1], [2].

Classification Results

The Classification Ablation Ladder

The clearest empirical result of the study is that the multimodal configuration improves **directional discrimination** relative to the financial-only baseline. Table 1 reports the best classification result for each feature configuration. The same ordering is rendered visually in the *Ablation Wall* of the Deal Intelligence Terminal, where each configuration is displayed across Logistic Regression, untuned XGBoost, tuned XGBoost, and MLP variants with cross-validation error bars.

Config	Description	Features	AUC-ROC	Accuracy	F1
M1	Financial only	56	0.5408 $\pm$ 0.013	52.8%	0.473
M2	Financial + aggregated text	184	0.5289 $\pm$ 0.027	52.9%	0.476
M3	Financial + text + graph	248	0.5655 $\pm$ 0.020	54.8%	0.490
M3e	M3 + engineered auxiliary features	261	0.5585 $\pm$ 0.016	55.1%	0.492

Table 1: Classification ablation ladder — best model result per feature configuration under five-fold stratified cross-validation. Bold indicates the headline AUC result.

The headline result is straightforward. M3 achieves AUC-ROC = **0.5655**, improving on the financial-only baseline M1 (0.5408) by **+0.0247** and on M2 (0.5289) by **+0.0366**. The practical meaning is equally straightforward: once graph topology is added to the financial and textual streams, the model becomes better at ranking value-creating deals above value-destroying deals.

A small but important clarification is needed for M3e. The engineered-feature extension increases accuracy slightly (55.1%) and F1 slightly (0.492), but does not improve AUC beyond M3. Because AUC is the primary classification metric defined in Chapter 3, M3 remains the headline classification model.

Hyperparameter Tuning: An Honest Negative Result

Hyperparameter search did not improve the XGBoost classifier. Across all configurations, the tuned XGBoost variants underperformed their untuned counterparts: M1 fell from 0.5408 to 0.5351, M2 from 0.5289 to 0.5252, and M3 from 0.5655 to 0.5555. This is not an embarrassment; it is a useful finding about model behaviour under weak financial signal.

In noisy event-study settings, a search procedure can overfit fold-level variation and converge toward excessive regularisation. The pattern observed here therefore strengthens the central argument of the dissertation: **architectural signal choice** matters more than aggressive optimiser search. The durable gain comes from adding graph information, not from squeezing the same financial-only feature space harder [3], [4].

Regression Results

Continuous CAR Remains Structurally Difficult

The regression pipeline was retained because Chapter 3 defined it as necessary for testing whether structural information can explain CAR magnitude, not merely direction. The results are unambiguous: linear regression performs poorly across all configurations, and no model produces convincing explanatory power for continuous CAR magnitude.

Config	De- scrip- tion	R^2	Interpretation
M1	Fi- nan- cial only	-0.008	No explanatory power above predicting the mean.
M2	Fi- nan- cial + ag- gre- gated text	-0.155	Text aggregation increases noise in continuous prediction.
M3	Fi- nan- cial + text + graph	-0.164	Multimodal fusion improves classification, not magnitude regression.

Table 2: Regression pipeline summary — representative continuous CAR results. Negative R^2 indicates that the model performs worse than predicting the sample mean.

These negative R^2 values are not evidence that the project failed. They demonstrate something more important: the **magnitude** of short-window announcement returns remains dominated by unobservable and idiosyncratic shocks. Payment method, takeover speculation, competing bids, macro conditions, investor sentiment, and timing noise all influence realised CAR in ways that are only partially visible in pre-announcement features [5], [6].

The regression findings therefore justify the chapter’s structure. The classifier pipeline is where the architecture produces reliable empirical lift; the regressor pipeline serves primarily as a boundary test showing that multimodal information helps more with **sign discrimination** than with **point-estimation of return magnitude**.

The M2 Reversal

Why Naive NLP Makes the Model Worse

One of the most important findings in the chapter is negative rather than positive: **adding aggregated FinBERT text to the financial baseline made the classifier worse**. M2 falls from 0.5408 to 0.5289, a drop of **-0.0119** AUC. This is not a rounding fluctuation. It is a directional reversal, and it explains why the text pipeline had to be designed carefully in Chapter 3.

The mechanism is conceptual and empirical at the same time. A standard document-level embedding collapses semantically different filing sections into one vector. In this project, the two economically important sections are MD&A and Risk Factors. MD&A language tends to encode

strategic coherence, managerial confidence, and integration ambition; Risk Factor language encodes concentration risk, regulatory exposure, and vulnerability. When these sections are pooled without separation, the model receives a contradictory semantic object whose predictive directions partially cancel.

The significance of M2 is therefore larger than its raw score suggests. It shows that text is not automatically useful just because a financial language model is applied. In M&A prediction, **section semantics matter**. The dissertation’s text contribution is not “FinBERT helps”; it is that undifferentiated text can hurt, while properly separated text can be made economically interpretable [7], [8], [9].

Hypothesis Tests

H1 — Topological Alpha (Supported)

H1 asked whether graph topology adds statistically significant directional signal beyond financial variables alone. On the classification pipeline, the answer is yes. M3 improves AUC-ROC over M1 by **+0.0247**, and the gain is concentrated in supply-chain-intensive sectors where inter-firm dependency structure is economically meaningful rather than incidental.

The substantive interpretation is direct. Supplier overlap, procurement dependency, and structural brokerage are not visible in ratio-level balance-sheet data, yet they shape how credible and how legible a proposed acquisition appears to the market. Where the network is economically dense, the graph modality captures information about integration plausibility that tabular finance alone cannot recover [10], [11], [12].

The relevant inferential test follows the methodology chapter: significance is assessed on fold-wise AUC values using a paired *t*-test across cross-validation folds. The directional result and the observed AUC gap support the claim that topology contributes information orthogonal to financial fundamentals. **Verdict: H1 is supported.**

H2 — Semantic Divergence (Supported)

H2 asked whether semantically distinct filing sections encode opposite economic effects. The OLS evidence supports that claim. On the semantic-divergence sample of $n = 1\{, \}$ 140 deals, the estimated coefficients are:

$$\beta_{\text{MDA}} = +0.0044 \quad \beta_{\text{RF}} = -0.0080 \quad R^2 = 0.0015$$

The signs are exactly as predicted. Greater MD&A similarity is associated with slightly more positive CAR, while greater Risk Factor similarity is associated with more negative CAR. The economic logic is intuitive: strategic similarity can signal integration coherence, but shared risk exposure can imply concentration rather than diversification [9], [13].

The low R^2 should be interpreted carefully rather than defensively. In event-study settings, a single semantic factor is not expected to explain a large proportion of return variance because announcement-period returns are jointly shaped by multiple concurrent forces [1]. The relevant test here is not whether text alone explains most of CAR, but whether the **direction** of the semantic effects matches theory. It does. Combined with the M2 reversal, the result supports the deeper claim that semantic aggregation obscures economically distinct textual mechanisms.

Verdict: H2 is supported.

H3 — Information Transparency Dampening (Supported)

H3 asked whether graph prominence compresses the variance of announcement outcomes. This is the strongest formally tested result in the chapter. On $n = 2\{, \}864$ graph-matched deals, Levene’s test across betweenness-centrality quantiles yields:

$$F_{\text{Levene}} = 7.0745 \quad p = 0.0079$$

The null of equal variance is rejected. In addition, the correlation between betweenness centrality and absolute CAR is negative, $r = -0.0701$ with $p = 0.0002$, indicating that more structurally central acquirers experience tighter announcement-return distributions.

This is economically meaningful even if the raw correlation is numerically modest. Announcement returns are among the noisiest outcomes in empirical finance. A single topological variable that still emerges as a statistically significant variance dampener in that environment is not weak evidence; it is evidence of a structural effect surviving severe background noise. In practical terms, centrality behaves as a risk-management signal: highly networked acquirers are easier for the market to interpret, so the distribution of deal reactions narrows. **Verdict: H3 is supported.**

Interpretability

SHAP and Economic Credibility

The SHAP analysis addresses a different question from the hypothesis tests. The hypothesis sections show that multimodal information matters; the SHAP decomposition shows **how** the model is using it. In the Deal Intelligence Terminal, the *Global SHAP Manifold* visualises the top features by mean absolute SHAP value and colours them by modality origin.

The result is economically reassuring. Traditional financial features still dominate the very top of the ranking, which is exactly what the corporate finance literature would predict. But graph embedding components and PCA-compressed text components also appear among the strongest contributors, and their SHAP contributions are not uniformly close to zero. That pattern matters because it shows the multimodal lift is not a cross-validation accident. The model is extracting real, non-trivial signal from graph and text features that linear tabular models would either ignore or fail to combine properly [14], [15].

The text-side SHAP pattern also aligns with H2. Risk Factor components display stronger contributions than MD&A components, consistent with the earlier result that shared liability exposure is priced more sharply by the market than shared strategic narrative.

Practical Meaning

Why a +0.025 AUC Gain Matters

A +0.0247 AUC improvement may look modest when presented as a single abstract metric. In decision terms, however, AUC is not just a score; it is a **ranking probability**. An AUC of 0.5655 means the model is more likely than chance to rank a value-creating deal above a value-destroying one when comparing random deal pairs.

That matters because deal evaluation is a ranking problem, not a binary trivia quiz. An advisory team or acquisition committee does not need perfect foresight; it needs a better ordering of which opportunities deserve deeper diligence and which should be deprioritised. A persistent 2.5 percentage-point improvement in pairwise ranking, applied across a large deal pipeline,

translates into systematically better attention allocation and lower probability of committing scarce time and capital to the worst opportunities [4], [16].

The contribution should therefore be stated carefully. The model does not “solve” M&A prediction. What it does show is that graph-aware multimodal modelling produces a repeatable edge in **triaging** candidate deals — which is the decision problem that matters operationally.

Limitations

Boundaries of the Findings

The findings should be interpreted within three clear limits. First, the headline AUC of 0.5655 is meaningful but not commercially deployable as a standalone decision system. The contribution of the chapter is the existence of multimodal signal lift, not the construction of a near-perfect forecasting engine [17].

Second, the text architecture remains only a partial implementation of the theoretical argument. Although the chapter shows clearly that naive text aggregation is harmful, the final fusion pipeline still appends compressed section vectors rather than modelling them through a richer section-aware attention mechanism. A more explicit dual-stream text encoder may recover additional semantic signal [18].

Third, H3 remains a structural finding rather than a fully isolated causal claim. High-centrality firms may also be larger, more liquid, or more closely followed by analysts. The chapter therefore interprets centrality as a statistically meaningful dampening correlate, not as an exhaustively isolated causal driver [12].

Synthesis

Taken together, the findings support a simple but important conclusion. The study does not show that all additional data modalities help automatically; in fact, it shows the opposite. Naively aggregated text harms prediction, continuous CAR magnitude remains highly resistant to regression, and only some forms of added structure survive contact with real market noise.

What **does** survive is economically structured multimodality. Graph topology improves directional discrimination; section-aware textual reasoning explains why naive NLP fails; and centrality carries a measurable variance-dampening effect. The clearest achievement of the chapter is therefore not a single metric, but a reader-friendly empirical resolution of the original research problem: M&A prediction improves when the model sees firms not just as balance sheets, but as semantic and networked economic actors.

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